

Newman-Penrose Formalism Calculations

Stephani et al., Equation (21.20)

Metric

$$\begin{aligned} ds^2 = & \left(-\frac{6}{6(e^2 + g^2)p^6 - 12mp^5 + 6ep^4 - 12lp^3 + (\Lambda - 6\gamma)p^2 + (6(e^2 + g^2)p^4 - 12mp^3 + 6ep^2 - 12lp + \Lambda - 6\gamma)q^2 + 2(6(e^2 + g^2)p^5 - 12mp^4 + 6ep^3 - 12lp^2 + (\Lambda - 6\gamma)p)q} \right) dp \\ & \otimes dp + \left(\frac{6}{6(e^2 + g^2)q^6 + 12((e^2 + g^2)p + m)q^5 + 6((e^2 + g^2)p^2 + 4mp + e)q^4 + 12(mp^2 + ep + l)q^3 - (\Lambda + 6\gamma)p^2 + (6ep^2 + 24lp - \Lambda - 6\gamma)q^2 + 2(6lp^2 - (\Lambda + 6\gamma)p)q} \right) dq \\ & \otimes dq + \left(-\frac{6(e^2 + g^2)p^4 - 12mp^3 + 6ep^2 - 12lp + \Lambda - 6\gamma}{6(p^2 + 2pq + q^2)} \right) d\sigma \otimes d\sigma + \left(-\frac{6(e^2 + g^2)q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}{6(p^2 + 2pq + q^2)} \right) d\tau \otimes d\tau \end{aligned}$$

Coordinates

$$(x^\mu) = (p, q, \sigma, \tau)$$

Metric Functions

$$X = -e^2 p^4 - g^2 p^4 + 2mp^3 - ep^2 + 2lp - \frac{1}{6}\Lambda + \gamma$$

$$Y = e^2 q^4 + g^2 q^4 + 2mq^3 + eq^2 + 2lq - \frac{1}{6}\Lambda - \gamma$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}}{2\sqrt{e^2 q^4 + g^2 q^4 + 2mq^3 + eq^2 + 2lq - \frac{1}{6}\Lambda - \gamma(p+q)}} \right) dq + \left(-\frac{\sqrt{2}\sqrt{e^2 q^4 + g^2 q^4 + 2mq^3 + eq^2 + 2lq - \frac{1}{6}\Lambda - \gamma}}{2(p+q)} \right) d\tau$$

$$n = \left(-\frac{\sqrt{2}}{2\sqrt{e^2q^4 + g^2q^4 + 2mq^3 + eq^2 + 2lq - \frac{1}{6}\Lambda - \gamma(p+q)}} \right) dq + \left(-\frac{\sqrt{2}\sqrt{e^2q^4 + g^2q^4 + 2mq^3 + eq^2 + 2lq - \frac{1}{6}\Lambda - \gamma}}{2(p+q)} \right) d\tau$$

$$m = \left(\frac{\sqrt{2}}{2\sqrt{-e^2p^4 - g^2p^4 + 2mp^3 - ep^2 + 2lp - \frac{1}{6}\Lambda + \gamma(p+q)}} \right) dp + \left(\frac{i\sqrt{2}\sqrt{-e^2p^4 - g^2p^4 + 2mp^3 - ep^2 + 2lp - \frac{1}{6}\Lambda + \gamma}}{2(p+q)} \right) d\sigma$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2\sqrt{-e^2p^4 - g^2p^4 + 2mp^3 - ep^2 + 2lp - \frac{1}{6}\Lambda + \gamma(p+q)}} \right) dp + \left(-\frac{i\sqrt{2}\sqrt{-e^2p^4 - g^2p^4 + 2mp^3 - ep^2 + 2lp - \frac{1}{6}\Lambda + \gamma}}{2(p+q)} \right) d\sigma$$

Contravariant components

$$\begin{aligned} l = & \left(\frac{\sqrt{3}e^2pq^4}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}g^2pq^4}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}e^2q^5}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \right. \\ & + \frac{\sqrt{3}g^2q^5}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{2\sqrt{3}mpq^3}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{2\sqrt{3}mq^4}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & + \frac{\sqrt{3}epq^2}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}eq^3}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{2\sqrt{3}lpq}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & + \frac{2\sqrt{3}lq^2}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} - \frac{\sqrt{3}\Lambda p}{6\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} - \frac{\sqrt{3}\gamma p}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & \left. - \frac{\sqrt{3}\Lambda q}{6\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} - \frac{\sqrt{3}\gamma q}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \right) dq \\ & + \left(\frac{\sqrt{3}p}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}q}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \right) d\tau \end{aligned}$$

$$\begin{aligned}
n = & \left(-\frac{\sqrt{3}e^2pq^4}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{\sqrt{3}g^2pq^4}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{\sqrt{3}e^2q^5}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \right. \\
& - \frac{\sqrt{3}g^2q^5}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{2\sqrt{3}mpq^3}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{2\sqrt{3}mq^4}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \\
& - \frac{\sqrt{3}epq^2}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{\sqrt{3}eq^3}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{2\sqrt{3}lpq}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \\
& - \frac{2\sqrt{3}lq^2}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} + \frac{\sqrt{3}\Lambda p}{6\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} + \frac{\sqrt{3}\gamma p}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \\
& \left. + \frac{\sqrt{3}\Lambda q}{6\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} + \frac{\sqrt{3}\gamma q}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \right) dq \\
& + \left(\frac{\sqrt{3}p}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} + \frac{\sqrt{3}q}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \right) d\tau \\
m = & \left(-\frac{\sqrt{3}e^2p^5}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} - \frac{\sqrt{3}g^2p^5}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \right. \\
& - \frac{\sqrt{3}e^2p^4q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} - \frac{\sqrt{3}g^2p^4q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& + \frac{2\sqrt{3}mp^4}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{2\sqrt{3}mp^3q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} - \frac{\sqrt{3}ep^3}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& - \frac{\sqrt{3}ep^2q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{2\sqrt{3}lp^2}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{2\sqrt{3}lpq}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& - \frac{\sqrt{3}\Lambda p}{6\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{\sqrt{3}\gamma p}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& - \frac{\sqrt{3}\Lambda q}{6\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{\sqrt{3}\gamma q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \left. \right) dp \\
& + \left(-\frac{i\sqrt{3}\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}p}{6e^2p^4+6g^2p^4-12mp^3+6ep^2-12lp+\Lambda-6\gamma} - \frac{i\sqrt{3}\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}q}{6e^2p^4+6g^2p^4-12mp^3+6ep^2-12lp+\Lambda-6\gamma} \right) d\sigma
\end{aligned}$$

$$\begin{aligned}
\overline{m} = & \left(-\frac{\sqrt{3}e^2p^5}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}g^2p^5}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \right. \\
& - \frac{\sqrt{3}e^2p^4q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}g^2p^4q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& + \frac{2\sqrt{3}mp^4}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{2\sqrt{3}mp^3q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}ep^3}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}ep^2q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{2\sqrt{3}lp^2}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{2\sqrt{3}lpq}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}\Lambda p}{6\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}\gamma p}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& \left. - \frac{\sqrt{3}\Lambda q}{6\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}\gamma q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \right) dp \\
& + \left(\frac{i\sqrt{3}\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}p}{6e^2p^4 + 6g^2p^4 - 12mp^3 + 6ep^2 - 12lp + \Lambda - 6\gamma} + \frac{i\sqrt{3}\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}q}{6e^2p^4 + 6g^2p^4 - 12mp^3 + 6ep^2 - 12lp + \Lambda - 6\gamma} \right) d\sigma
\end{aligned}$$

Directional Derivatives

$$\begin{aligned}
D = & \frac{\sqrt{3}e^2pq^4}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}g^2pq^4}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}e^2q^5}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\
& + \frac{\sqrt{3}g^2q^5}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{2\sqrt{3}mpq^3}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{2\sqrt{3}mq^4}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\
& + \frac{\sqrt{3}epq^2}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}eq^3}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{2\sqrt{3}lpq}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\
& + \frac{2\sqrt{3}lq^2}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} - \frac{\sqrt{3}\Lambda p}{6\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} - \frac{\sqrt{3}\gamma p}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\
& - \frac{\sqrt{3}\Lambda q}{6\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} - \frac{\sqrt{3}\gamma q}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \partial_q \\
& + \frac{\sqrt{3}p}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}q}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \partial_\tau
\end{aligned}$$

$$\begin{aligned}
\Delta = & -\frac{\sqrt{3}e^2pq^4}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{\sqrt{3}g^2pq^4}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{\sqrt{3}e^2q^5}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \\
& - \frac{\sqrt{3}g^2q^5}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{2\sqrt{3}mpq^3}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{2\sqrt{3}mq^4}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \\
& - \frac{\sqrt{3}epq^2}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{\sqrt{3}eq^3}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} - \frac{2\sqrt{3}lpq}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \\
& - \frac{2\sqrt{3}lq^2}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} + \frac{\sqrt{3}\Lambda p}{6\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} + \frac{\sqrt{3}\gamma p}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \\
& + \frac{\sqrt{3}\Lambda q}{6\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} + \frac{\sqrt{3}\gamma q}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \partial_q \\
& + \frac{\sqrt{3}p}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} + \frac{\sqrt{3}q}{\sqrt{6e^2q^4+6g^2q^4+12mq^3+6eq^2+12lq-\Lambda-6\gamma}} \partial_\tau
\end{aligned}$$

$$\begin{aligned}
\delta = & -\frac{\sqrt{3}e^2p^5}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} - \frac{\sqrt{3}g^2p^5}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} - \frac{\sqrt{3}e^2p^4q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& - \frac{\sqrt{3}g^2p^4q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{2\sqrt{3}mp^4}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& + \frac{2\sqrt{3}mp^3q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} - \frac{\sqrt{3}ep^3}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} - \frac{\sqrt{3}ep^2q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& + \frac{2\sqrt{3}lp^2}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{2\sqrt{3}lpq}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& - \frac{\sqrt{3}\Lambda p}{6\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{\sqrt{3}\gamma p}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \\
& - \frac{\sqrt{3}\Lambda q}{6\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} + \frac{\sqrt{3}\gamma q}{\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}} \partial_p \\
& - \frac{i\sqrt{3}\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}p}{6e^2p^4+6g^2p^4-12mp^3+6ep^2-12lp+\Lambda-6\gamma} - \frac{i\sqrt{3}\sqrt{-6e^2p^4-6g^2p^4+12mp^3-6ep^2+12lp-\Lambda+6\gamma}q}{6e^2p^4+6g^2p^4-12mp^3+6ep^2-12lp+\Lambda-6\gamma} \partial_\sigma
\end{aligned}$$

$$\begin{aligned}
\bar{\delta} = & -\frac{\sqrt{3}e^2p^5}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}g^2p^5}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}e^2p^4q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}g^2p^4q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{2\sqrt{3}mp^4}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& + \frac{2\sqrt{3}mp^3q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}ep^3}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}ep^2q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& + \frac{2\sqrt{3}lp^2}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{2\sqrt{3}lpq}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}\Lambda p}{6\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}\gamma p}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}\Lambda q}{6\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}\gamma q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \partial_p \\
& + \frac{i\sqrt{3}\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}p}{6e^2p^4 + 6g^2p^4 - 12mp^3 + 6ep^2 - 12lp + \Lambda - 6\gamma} + \frac{i\sqrt{3}\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}q}{6e^2p^4 + 6g^2p^4 - 12mp^3 + 6ep^2 - 12lp + \Lambda - 6\gamma} \partial_\sigma
\end{aligned}$$

Spin Coefficients

$$\rho = \frac{1}{6} \sqrt{3} \sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{6} \sqrt{3} \sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\begin{aligned}
\alpha = & \frac{\sqrt{3}e^2p^4}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}g^2p^4}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}e^2p^3q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& + \frac{\sqrt{3}g^2p^3q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}mp^3}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{3\sqrt{3}mp^2q}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}epq}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& + \frac{\sqrt{3}lp}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}lq}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}\Lambda}{12\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}\gamma}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}}
\end{aligned}$$

$$\begin{aligned}
\beta = & -\frac{\sqrt{3}e^2p^4}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}g^2p^4}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}e^2p^3q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}g^2p^3q}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& + \frac{\sqrt{3}mp^3}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{3\sqrt{3}mp^2q}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}epq}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}lp}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& + \frac{\sqrt{3}lq}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& + \frac{\sqrt{3}\Lambda}{12\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}\gamma}{2\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}}
\end{aligned}$$

$$\begin{aligned}
\tau = & -\frac{\sqrt{3}e^2p^4}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}g^2p^4}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{2\sqrt{3}mp^3}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}ep^2}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{2\sqrt{3}lp}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\
& - \frac{\sqrt{3}\Lambda}{6\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}\gamma}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}}
\end{aligned}$$

$$\begin{aligned}\pi = & \frac{\sqrt{3}e^2p^4}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} + \frac{\sqrt{3}g^2p^4}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{2\sqrt{3}mp^3}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\ & + \frac{\sqrt{3}ep^2}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{2\sqrt{3}lp}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} \\ & + \frac{\sqrt{3}\Lambda}{6\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}} - \frac{\sqrt{3}\gamma}{\sqrt{-6e^2p^4 - 6g^2p^4 + 12mp^3 - 6ep^2 + 12lp - \Lambda + 6\gamma}}\end{aligned}$$

$$\begin{aligned}\epsilon = & \frac{\sqrt{3}e^2pq^3}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}g^2pq^3}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}e^2q^4}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & + \frac{\sqrt{3}g^2q^4}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{3\sqrt{3}mpq^2}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}mq^3}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & + \frac{\sqrt{3}epq}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}lp}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} - \frac{\sqrt{3}lq}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & + \frac{\sqrt{3}\Lambda}{12\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}\gamma}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}}\end{aligned}$$

$$\begin{aligned}\gamma = & \frac{\sqrt{3}e^2pq^3}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}g^2pq^3}{\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}e^2q^4}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & + \frac{\sqrt{3}g^2q^4}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{3\sqrt{3}mpq^2}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}mq^3}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & + \frac{\sqrt{3}epq}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}lp}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} - \frac{\sqrt{3}lq}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} \\ & + \frac{\sqrt{3}\Lambda}{12\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}} + \frac{\sqrt{3}\gamma}{2\sqrt{6e^2q^4 + 6g^2q^4 + 12mq^3 + 6eq^2 + 12lq - \Lambda - 6\gamma}}\end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{2} e^2 p^4 + \frac{1}{2} g^2 p^4 + 2 e^2 p^3 q + 2 g^2 p^3 q + 3 e^2 p^2 q^2 + 3 g^2 p^2 q^2 + 2 e^2 p q^3 + 2 g^2 p q^3 + \frac{1}{2} e^2 q^4 + \frac{1}{2} g^2 q^4$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = \frac{1}{6} \Lambda$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -e^2 p^4 - g^2 p^4 - 2 e^2 p^3 q - 2 g^2 p^3 q + 2 e^2 p q^3 + 2 g^2 p q^3 + e^2 q^4 + g^2 q^4 + m p^3 + 3 m p^2 q + 3 m p q^2 + m q^3$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "